

A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

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SOICT 2026

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Good morning. I will show what a second edge board actually buys you when you split an ML intrusion detector across two Jetson Orin Nano Super devices. Seven minutes, three results.

I. PROBLEM AND CONTRIBUTIONS

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Edge intrusion detection: three open gaps

Detectors now *fit* on single-board hardware — but a monolithic one classifies **every** flow and saturates an 8 GB ARM board.

What the literature does not tell us

- 1. Everything is measured on a **single device**.
- 2. Throughput, latency and **energy** are never reported together.
- 3. Offline scores come from **leaky protocols**.

Our question

How should inference be *placed* across a constrained two-node edge cluster?

Our answer

Split it: a cheap **autoencoder gate** on board #1, the heavy **PySpark classifier** on board #2, joined by a Kafka topic.

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I. PROBLEM AND CONTRIBUTIONS

└ Edge intrusion detection: three open gaps

Three gaps: single-device evaluation, energy never reported next to latency, and offline numbers taken from leaky pipelines. Which model to deploy is the other half of the question. We attack all three with one system.

Edge intrusion detection: three open gaps

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Split it: a cheap **autoencoder gate** on board #1, the heavy **PySpark classifier** on board #2 joined by a Kafka topic.

1. A **two-stage distributed IDS**: anomaly gate + Spark classifier, physically separated across two boards.
2. **Three deployment modes** for a two-node Jetson cluster — pipeline split, horizontal Kafka scaling, Spark standalone cluster.
3. An **end-to-end evaluation**: throughput, latency percentiles, per-node CPU/RAM *and* energy per verdict.
4. **Leakage-aware provenance** — the deployed model is exactly the one our companion study selected under a leakage-free protocol.

2026-08-15

A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

I. PROBLEM AND CONTRIBUTIONS

Contributions

Four contributions. The fourth matters: the model we deploy is not a fresh model, it is the one that survived a strict offline audit.

1. A **two-stage distributed IDS**: anomaly gate + Spark classifier, physically separated across two boards.
2. **Three deployment modes** for a two-node Jetson cluster — pipeline split, horizontal Kafka scaling, Spark standalone cluster.
3. An **end-to-end evaluation**: throughput, latency percentiles, per-node CPU/RAM *and* energy per verdict.
4. **Leakage-aware provenance** — the deployed model is exactly the one our companion study selected under a leakage-free protocol.

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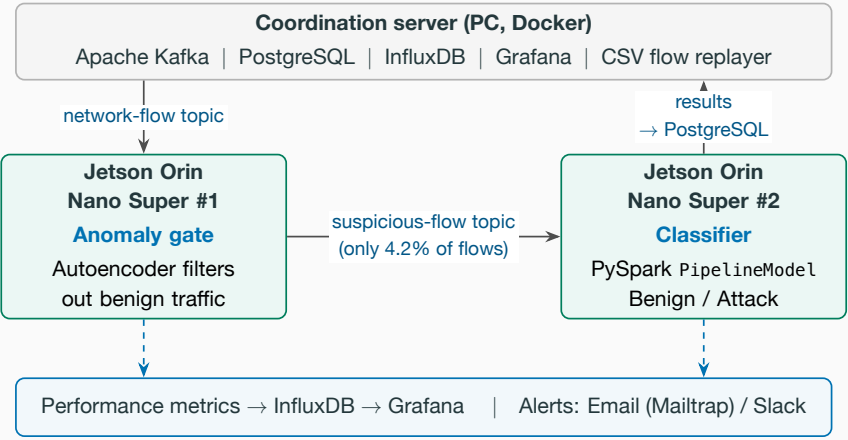
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└─ II. SYSTEM AND EVALUATION SETUP

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Split-deployment architecture



The gate's MSE threshold is calibrated *off/line* on a held-out benign validation split → committed before evaluation, never tuned on the test set.

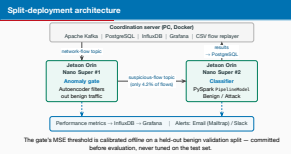
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II. SYSTEM AND EVALUATION SETUP

Split-deployment architecture

Host runs the infrastructure. Board one gates, board two classifies. Only suspicious flows cross the Kafka topic between them. Everything is instrumented into Grafana.



Three deployment modes, one workload

Single node full pipeline in one process — the reference.

A: Pipeline split gate on #1, classifier on #2 (our design).

B: Horizontal both boards run the *full* role, sharing a Kafka consumer group.

C: Spark cluster host as master, both boards as workers.

Throughput is counted as **final verdicts per second** — a flow is decided at the gate or at the classifier — so forwarded flows are never double-counted.

Setup

2× Jetson Orin Nano Super
(6-core A78AE, 8 GB, ≈67 TOPS)
PySpark 3.4.1, Kafka 3.5, Wi-Fi LAN
CICIDS2017 replay at 100 flows/s
≥3 repetitions, warmup excluded
Energy via tegrastats, idle subtracted

2026-08-15

A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

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Three deployment modes, one workload

Four configurations, identical workload. The verdict-per-second definition is what makes the modes comparable.

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A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

└ III. RESULTS AND CONCLUSION

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Result 1: splitting the pipeline buys 24× throughput

Mode	Throughput (flows/s)	p95 latency (ms)	Gate skip (%)	Energy/verdict (mJ)
Single node	2.60 ± 0.41	12.3 ± 3.0	(inline)	2490
A: Pipeline split	62.5 ± 0.6	13.1 ± 2.6	95.8	178
B: Horizontal	5.9 ± 1.4	21.6 ± 19.2	(inline)	2250
C: Spark cluster	90.0 ± 25.6	23.9 ± 17.8	97.9	119

- **A: 24× throughput at unchanged p95** — the gate absorbs 95.8% of benign traffic.
- **B stays Spark-bound** (2×): every node still classifies.
- C is fastest on average but ±26 flows/s and needs the host master.
- Energy amortised over 24–35× more verdicts: **2.5 → 0.12–0.18 J**.

Pipeline split is the recommended default — the highest throughput *per unit of variance*; cluster mode is for models that exceed single-node memory.

2026-08-15

A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

III. RESULTS AND CONCLUSION

Result 1: splitting the pipeline buys 24× throughput

This is the core table. Twenty-four times, at the same p95. Mode B proves the gain comes from role separation, not from adding a board. Energy is total-board per verdict: the active power delta was below the tegrastats noise floor at this load.

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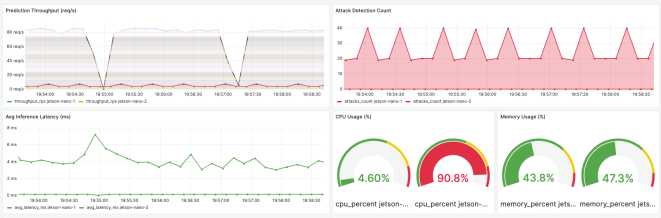
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Pipeline split is the recommended default — the highest throughput per unit of variance; cluster mode is for models that exceed single-node memory.

Result 2: the load asymmetry is visible live



At peak load, split mode:

- gate node: **4.6% CPU**
- classifier node: **90.8% CPU**

The classifier dominates the compute budget — which is exactly why moving it off the ingest path pays.

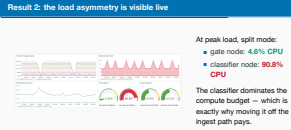
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III. RESULTS AND CONCLUSION

Result 2: the load asymmetry is visible live

Live Grafana at peak load. Five percent versus ninety percent. The asymmetry is the whole argument for the split. Add: the gate is a tunable filter, not a fixed point — backup slide has the operating curve.



Result 3: the honest price of one unified stack

We serve with Spark for consistency *with distributed training*. What does that cost? Same forest, same 29 deployed features, same board:

Engine	Throughput (flows/s)	p95 (ms)	RAM (%)	Energy/Inf. (mJ)
PySpark (deployed)	22.6	532	24.3	71.6
scikit-learn	204.4	49.8	13.3	2.91
ONNX Runtime	7870.8	1.1	15.7	0.06

ONNX Runtime sustains **348×** the deployed PySpark throughput. We report this as a *target for the next iteration* — export to ONNX/TensorRT for serving, keep Spark for training.

2026-08-15

A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

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Result 3: the honest price of one unified stack

We could have hidden this. Instead: the JVM costs us three orders of magnitude, and that sets the roadmap.

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What we showed

- A reproducible distributed edge IDS: Kafka + PySpark on a Jetson Orin Nano Super cluster.
- Splitting gate and classifier across boards: **2.6** → **62.5 flows/s** at p95 ≈13 ms, gate skipping 95.8%.
- Energy per verdict: **2.5 J** → **0.18 J**.

Limitations

- Two-node lab cluster over Wi-Fi; CICIDS2017 replay, not live traffic.
- The ≈67-TOPS GPU is unused by the CPU-only pipeline.
- Evasive traffic untested.

Next

- TensorRT/ONNX in the streaming path.
- Live traffic; federated retraining under drift.

Thank you — questions?

2026-08-15

A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

III. RESULTS AND CONCLUSION

Conclusion

Recap the three numbers: twenty-four times, thirteen milliseconds, ninety-six percent skipped. Then stop talking.

Conclusion	
What we showed ■ A reproducible distributed edge IDS: Kafka + PySpark on a Jetson Orin Nano Super cluster. ■ Splitting gate and classifier across boards: 2.6 → 62.5 flows/s at p95 ≈13 ms, gate skipping 95.8%. ■ Energy per verdict: 2.5 J → 0.18 J.	Limitations ■ Two-node lab cluster over Wi-Fi; CICIDS2017 replay, not live traffic. ■ The ≈67-TOPS GPU is unused by the CPU-only pipeline. ■ Evasive traffic untested. Next ■ TensorRT/ONNX in the streaming path. ■ Live traffic; federated retraining under drift. Thank you — questions?

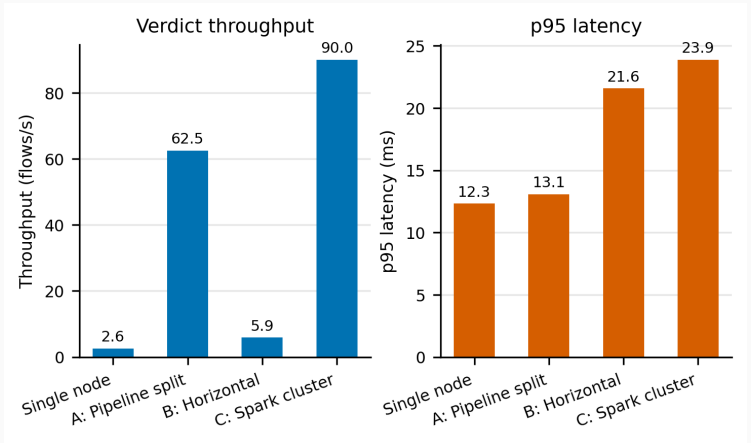
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Backup slides

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Backup: deployment-mode comparison chart

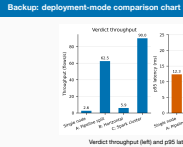


Verdict throughput (left) and p95 latency (right).

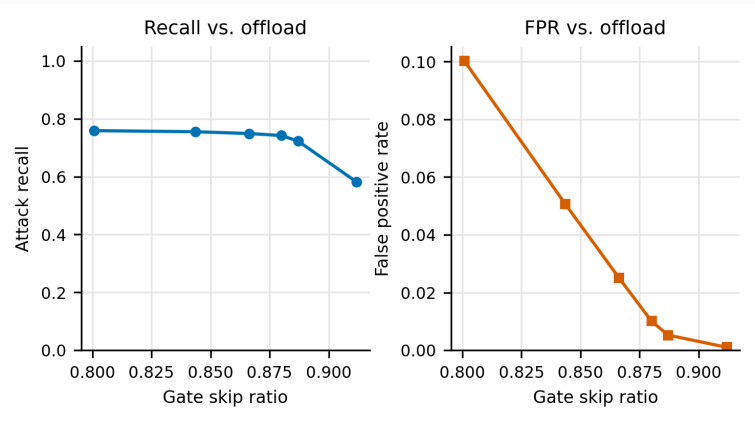
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Backup: deployment-mode comparison chart



Backup: the gate is an operating curve, not a point

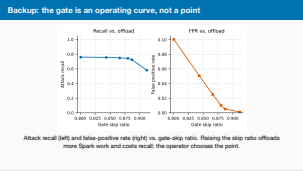


Attack recall (left) and false-positive rate (right) vs. gate-skip ratio. Raising the skip ratio offloads more Spark work and costs recall; the operator chooses the point.

2026-08-15

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Backup: the gate is an operating curve, not a point



Backup: which model, and why

Configuration	F1	Features	Size (MB)	Edge rationale
Baseline + XGBoost	0.9971	77	2.32	Reference accuracy
SHAP Top-30 + XGBoost	0.9970	30	0.003 [†]	Explainable; de- ployed set
RF Top-30 + XGBoost	0.9922	30	2.52	Embedded ranking
PCA k=40 + XGBoost	0.9939	40	3.82	Unsupervised reduc- tion

We deploy **SHAP Top-30 with Random Forest** (offline F1 0.9955): a native Spark ML estimator, so the exported PipelineModel loads on ARM64 without the native libraries Spark's XGBoost integration needs. The input vector has **29** features — destination_port is in the SHAP ranking but excluded as label-leaking.

[†] Distributed saving under-reports size; true size ≈2.5 MB.

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Backup: full monitoring dashboard



Per-node throughput, detected attacks, inference latency, CPU/RAM for both boards,
PostgreSQL-backed alerts.

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Backup: measurement protocol

- Host-side orchestration starts pipeline roles on the Jetsons over SSH, drives the load, collects per-node logs.
- ≥ 3 repetitions per configuration; warmup traffic *excluded* from the measured window (no JVM/JIT inflation of early batches).
- Metric queries aligned to the exact load window, not a trailing range.
- Latency percentiles per node over *raw per-flow* samples; cluster p95 is conservatively the *worst node's* p95.
- End-to-end send-to-verdict latency from producer timestamps under NTP-synchronised clocks (excludes DB-write cost).
- Energy: tegrastats, 30 s idle baseline then load window; two-node modes sum both boards.

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